ECE 310 Fall 2026
Lecture 1: Introduction to Digital Signals

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Welcome to ECE 310!

Let's take a look at the course website.



ECE 310 Frequently Asked Questions



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- Is ECE 310 harder than ECE 210?



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- Is ECE 310 harder than ECE 210?
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- Why should I care about ECE 310?



ECE 310 Frequently Asked Questions

- Is ECE 310 harder than ECE 210?
- I had a really tough time with ECE 210. Should I be concerned taking ECE 310?
- Why should I care about ECE 310?
- What questions do you have?



What is signal processing?

What are some words, phrases, or applications you associate with signal processing?



Learning Objectives

- Describe what defines a signal.
- Explain the difference between continuous-time and discrete-time signals.
- Understand the advantages of working with digital signals over analog signals.



What is a signal?



What is a signal?

- A signal is any measurable quantity that varies over time, space, or any other variable that orders or organizes these measurements.



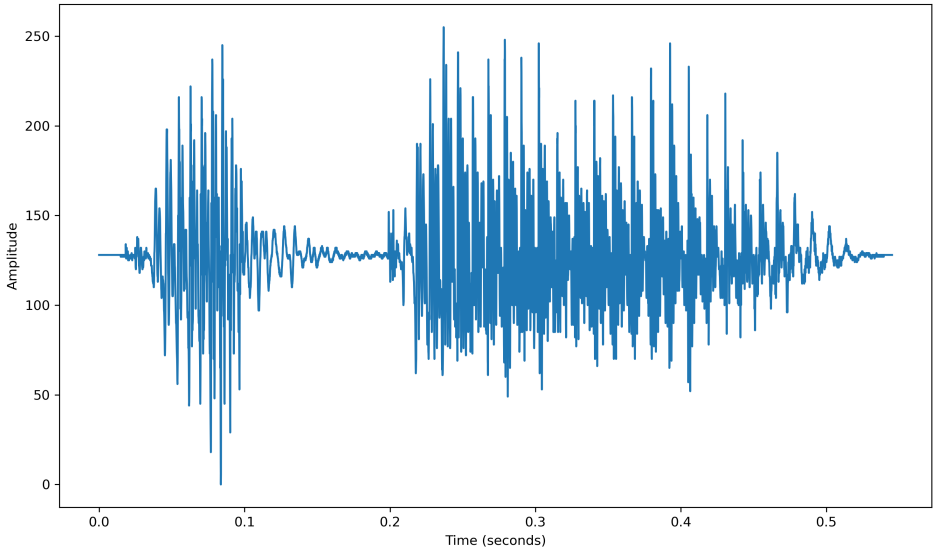
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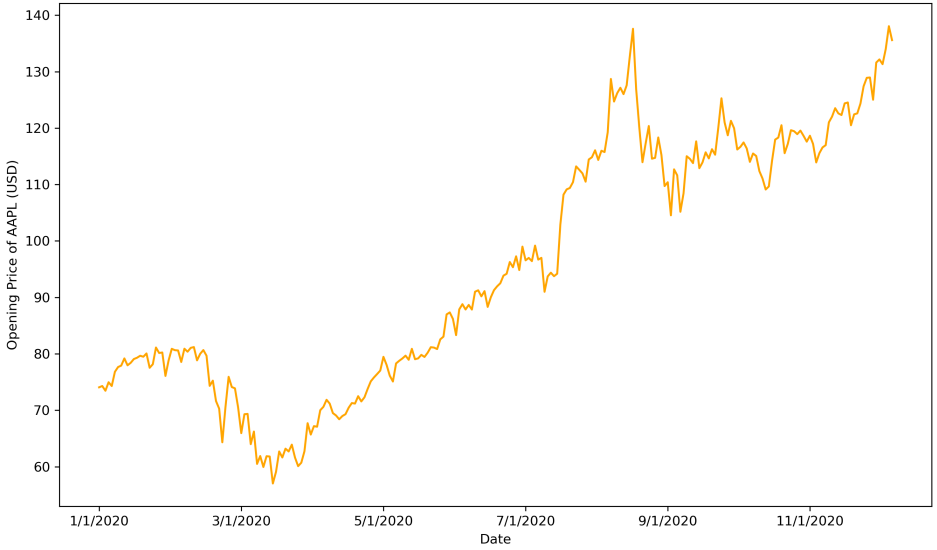
This sounds quite broad, but that's because signal processing is so widely applicable!



Signal Examples: Audio Signal



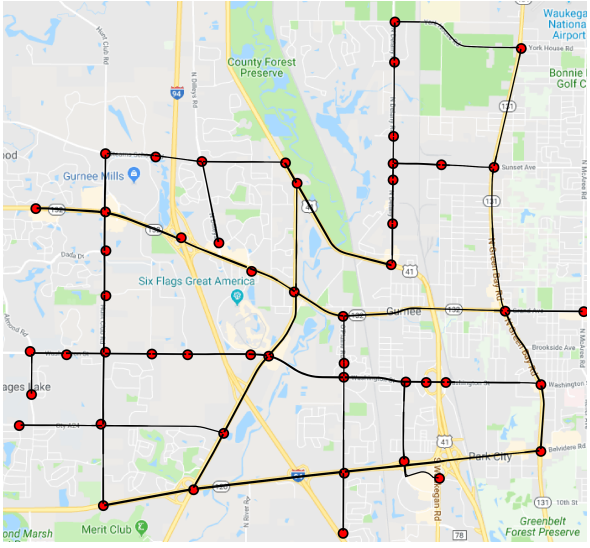
Signal Examples: Financial Signal



Signal Examples: Image Signal



Signal Examples: Graph Signal



Continuous-time vs. Discrete-time Signals

t is continuous

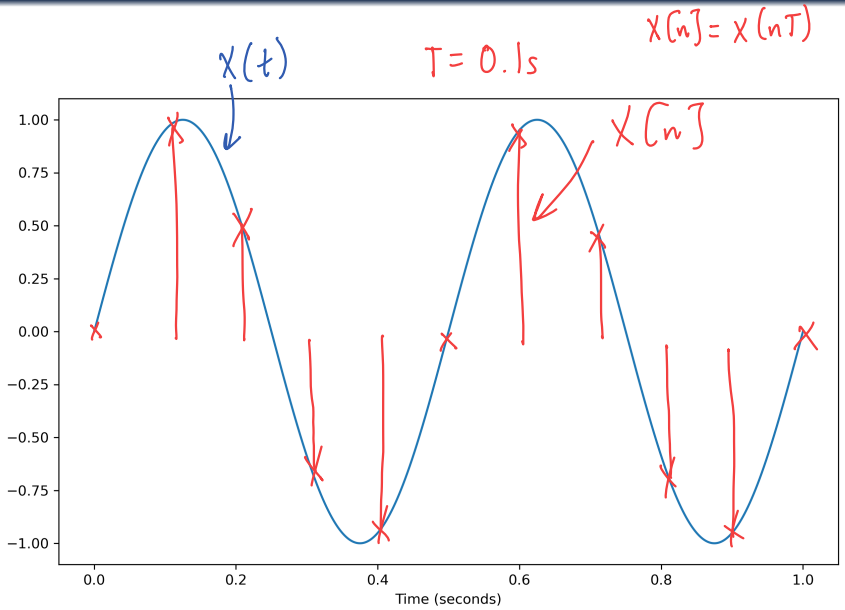
- A *continuous-time* signal $x(t)$ is observed continuously over time. Every time interval contains infinitely many values from the signal.
- A *discrete-time* signal $x[n]$ is observed at regularly spaced points in time. We say that the time-axis is *discretized*.
- For example, audio is commonly recorded at 44.1 kHz. Thus, we have a new sample every $T = 1/44{,}100$ seconds. T is called the *sampling period*.

$$x[n] = x(nT)$$

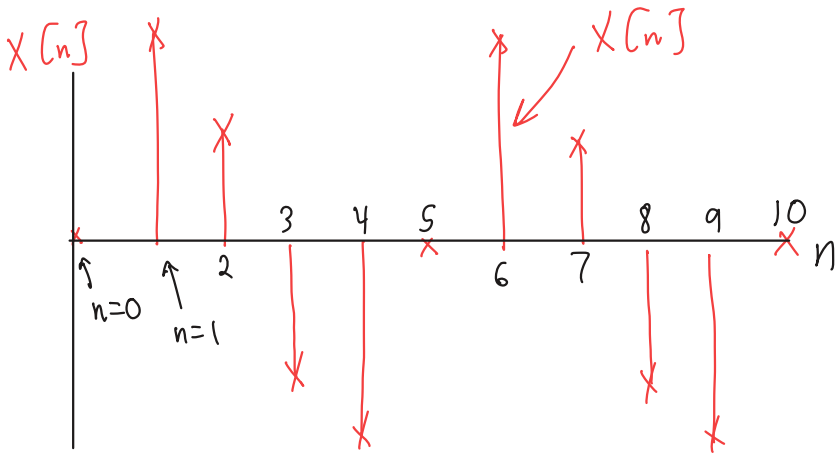
$n \in \mathbb{Z}$

integers, both positive or negative

Continuous-time vs. Discrete-time Signals



- $x[0] = x(0 \cdot 0.1)$
- $x[1] = x(1 \cdot 0.1)$
- $x[2] = x(2 \cdot 0.1)$
- ⋮


x[n]
x[n]
3
4
5
6
7
8
9
10
n
2
n=0
n=1

Continuous-valued vs. Discrete-valued Signals

- A *continuous-valued* signal takes any possible continuous value for its measurements.
- A *discrete-valued* signal takes one of a finite number of pre-defined values. The measurements are said to be *quantized*.



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- A discrete-time, discrete-valued signal is referred to as a *digital signal*.*



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- A discrete-time, discrete-valued signal is referred to as a *digital signal*.*
  - We will assume our signals are continuous-valued in this class since modern computing systems have very high precision.



Why digital signals?

What are some advantages of using digital signals over analog signals?

1. Math is much easier
   - ↳ Ex: $e^{x(t)}$ vs. $e^{x[n]}$
2. Digital electronics are more reliable
3. Cheap storage
4. Design flexibility